

MOHAMMAD SAMIR KHAN

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PROFESSIONAL SUMMARY

Backend Engineer with hands-on experience designing robust REST APIs, microservices, and high-performance server-side architectures using Python (FastAPI) and Node.js. Skilled in database modeling (PostgreSQL, MongoDB), custom JWT authentication layers, and implementing scalable containerized workflows via Docker. Proven ability to build full-stack content management pipelines and privacy-first background data ingestion trackers, while keeping a strong foundation in core engineering principles, concurrency, and data structures.

TECHNICAL SKILLS

Backend & Core:	Python, Node.js (Express), FastAPI, RESTful APIs, Microservices, Middleware Architecture, Authentication (JWT, bcrypt), Data Structures & Algorithms
Databases & Cloud:	PostgreSQL, MongoDB Atlas, Mongoose ODM, Caching concepts, Docker, Cloudinary CDN, Git, GitHub Actions (CI/CD), AWS (Familiar)
GenAI & Evaluation:	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), FAISS, ChromaDB, Fine-tuning (LoRA, QLoRA), Model Evaluation (SHAP, BLEU)

TECHNICAL PROJECTS

Full-Stack Dynamic CMS & Privacy-First Analytics Platform | *Next.js, Node.js, Express, MongoDB, TypeScript, Docker*

- Architected a decoupled full-stack CMS platform utilizing Next.js (App Router) and Express.js, reducing asset and content update deployment loops to zero without requiring manual server redeployments.
- Engineered a secure administrative custom framework with token-based JWT authentication, custom route middleware protection, bcrypt password encryption, and cryptographic recovery workflows.
- Built a self-hosted, privacy-first asynchronous web analytics ingestion tracker utilizing non-blocking browser background threads (navigator.sendBeacon) and geo-location APIs to deduplicate client metadata streams.
- Designed complex MongoDB Atlas aggregation pipelines (\$group, \$match, \$aggregate) to compute live 30-day traffic metrics, platform usage distributions, and unique visitor metrics displayed on a responsive dashboard.
- Integrated multi-format polymorphic asset streaming middleware via Multer and Cloudinary SDK, processing concurrent multi-part form data streams for high-speed automated file delivery.

AI Financial Advisor Workflow Engine | *Python, LangChain, Mistral, ChromaDB, ThreadPoolExecutor*

- Designed a tool-calling backend orchestration workflow combining deterministic rule-based database integrity checks with language model execution graphs to systematically prevent runtime factual hallucinations.
- Built a concurrent batch-processing engine using ThreadPoolExecutor to handle multi-threaded dataset parsing over 100+ structural records simultaneously, optimizing server context window constraints.
- Implemented rigorous data validation filters and structured verification quality gates across the application workflow layer, ensuring error isolation before transaction endpoints reached active client clients.

MediAI Healthcare Engine | *FastAPI, PostgreSQL, Llama 3.1, FAISS, XGBoost, SHAP*

- Developed a high-performance retrieval-augmented (RAG) backend query infrastructure using FastAPI and FAISS semantic indexing over complex unstructured multi-document medical data layers.
- Built a structured data extraction and optical parsing backend script using specialized fine-tuned transformer layers (BioBERT) to systematically normalize unformatted raw reports into indexed relational structures.

WORK EXPERIENCE

Software Engineer Intern | Trex InfoTech, Mumbai

May 2025 – Aug 2025

- Developed server-side RESTful API endpoints utilizing Node.js/Express and MongoDB backend layouts, strictly applying routing modularization, data structures validation, and unified HTTP error-handling standards.
- Authored an exhaustive multi-scenario Postman automation testing suite targeting integration flows, schema boundaries, and regression verification across production microservice clusters.
- Collaborated on agile repository management patterns and multi-stage branch merges via strict team Git-based workflows and centralized code review guidelines.

AI/ML Data Engineering Intern | Ybl Foundation, Mumbai

Jun 2024 – Jul 2024

- Designed multi-variable statistical data validation and hypothesis pipelines to isolate high-impact structural features from messy datasets using cross-validated scikit-learn models.
- Optimized data transformation queries and processing pipelines to accelerate exploration runtime bounds across internal development workflows.

EDUCATION

B.E., Computer Science (AI/ML Specialization) Viva Institute of Technology, Mumbai	Graduated: Jun 2026
Diploma, Computer Science & Engineering Pravin Patil Polytechnic, Mumbai	Graduated: Jun 2023